

Wrocław University of Science and Technology

Faculty of Management

Physics of complex systems

Report on cellular automata

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1 Wolfram Elementary Cellular Automata

1.1 Definition

In the work of Stephen Wolfram, elementary cellular automata are the simplest class of one-dimensional cellular automata, as they consist of a row of cells, each of which can be in one of two states (typically 0 or 1, or black and white). The system evolves in discrete time steps according to a fixed rule that updates every cell simultaneously, based only on its own current state and the states of its immediate left and right neighbors. Because each neighborhood of three cells has eight possible configurations and each configuration can map to either of two outcomes, there are 256 possible elementary rules, conventionally numbered using Wolfram's rule-numbering scheme, with famous examples such as Rule 30 and Rule 110 demonstrating how extremely simple local rules can generate behavior ranging from repetitive and predictable patterns to apparently random structures and even universal computation.

1.2 Discussion

“

- **Rule 30:** Why is it classified as Class III (chaotic)? How does the sequential update change the spacetime diagram?
- **Rule 110:** Why is it classified as Class IV (complex) ? Do the localised particle-like structures survive under sequential update? Why or why not?
- **General:** In 1-D ECA, which rule class benefits most from synchrony?

”

Rule 30 is classified as a Class III cellular automaton because its evolution produces highly chaotic and seemingly random patterns that do not settle into stable or repeating structures, even when initiated from a single active cell. Under sequential rather than synchronous updating, its characteristic triangular structures are disrupted, resulting in patterns that resemble scattered television static. In contrast, Rule 110 belongs to Class IV because it generates localized moving structures, often described as particles or propagating lines, which interact in complex ways across the grid. These emergent structures depend critically on synchronous updating, since their motion and interactions require all cells to update simultaneously in order to preserve consistent timing relationships. Moreover, Class IV cellular automata are especially sensitive to the choice of update scheme because their complex behavior arises from coordinated interactions among evolving structures, and asynchronous or sequential updates tend to disrupt these interactions and eliminate the organized dynamics that characterize computationally rich behavior.

1.3 Rule 30 results

Rule 30

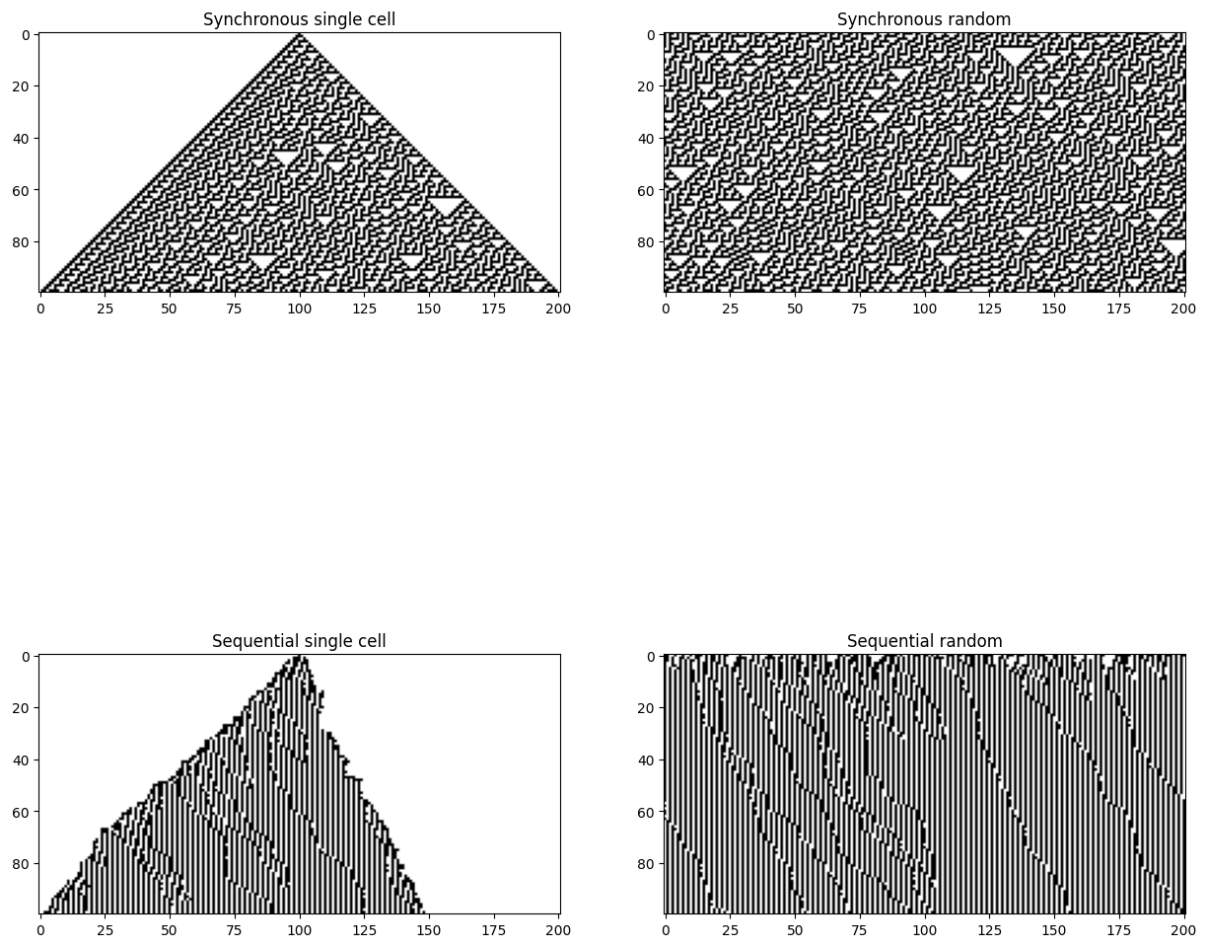


Figure 1: Rule 30 results

1.4 Rule 110 results

Rule 110

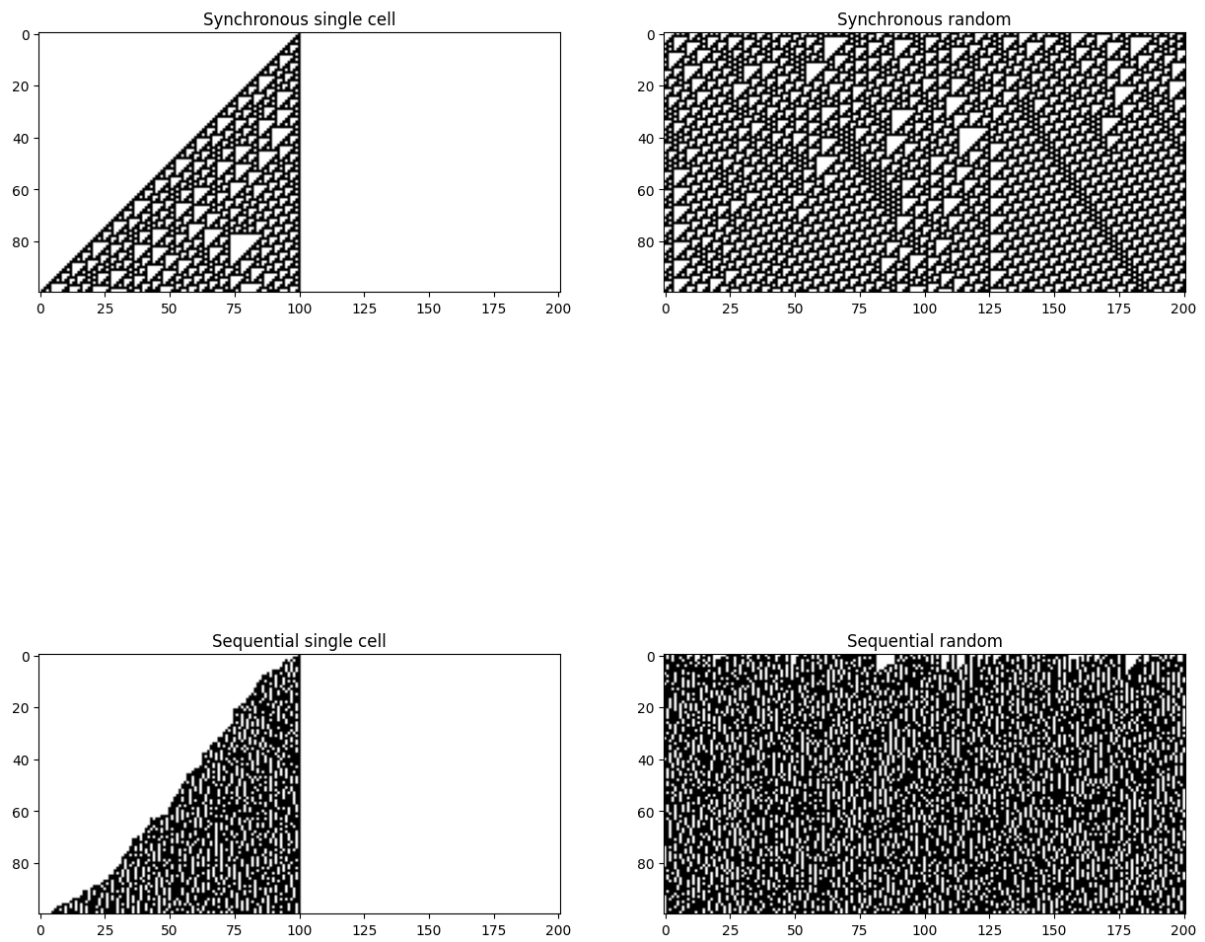


Figure 2: Rule 110 results

2 Conway's Game of Life

2.1 Definition

Conway's Game of Life is a two-dimensional cellular automaton devised by John Horton Conway in 1970 in which a grid of cells evolves over discrete time steps according to a small set of deterministic rules based on the states of neighboring cells. Each cell is either alive or dead, and at every generation, all cells update simultaneously. A living cell survives only if it has two or three living neighbors, a dead cell becomes alive if it has exactly three living neighbors, and all other cells either die or remain dead. Despite the simplicity of these local rules, the system is capable of producing highly complex emergent behavior, including stable structures, oscillators, and moving patterns such as gliders, making the Game of Life one of the most influential examples of how simple computational rules can generate unpredictable and computationally universal behavior.

2.2 Results

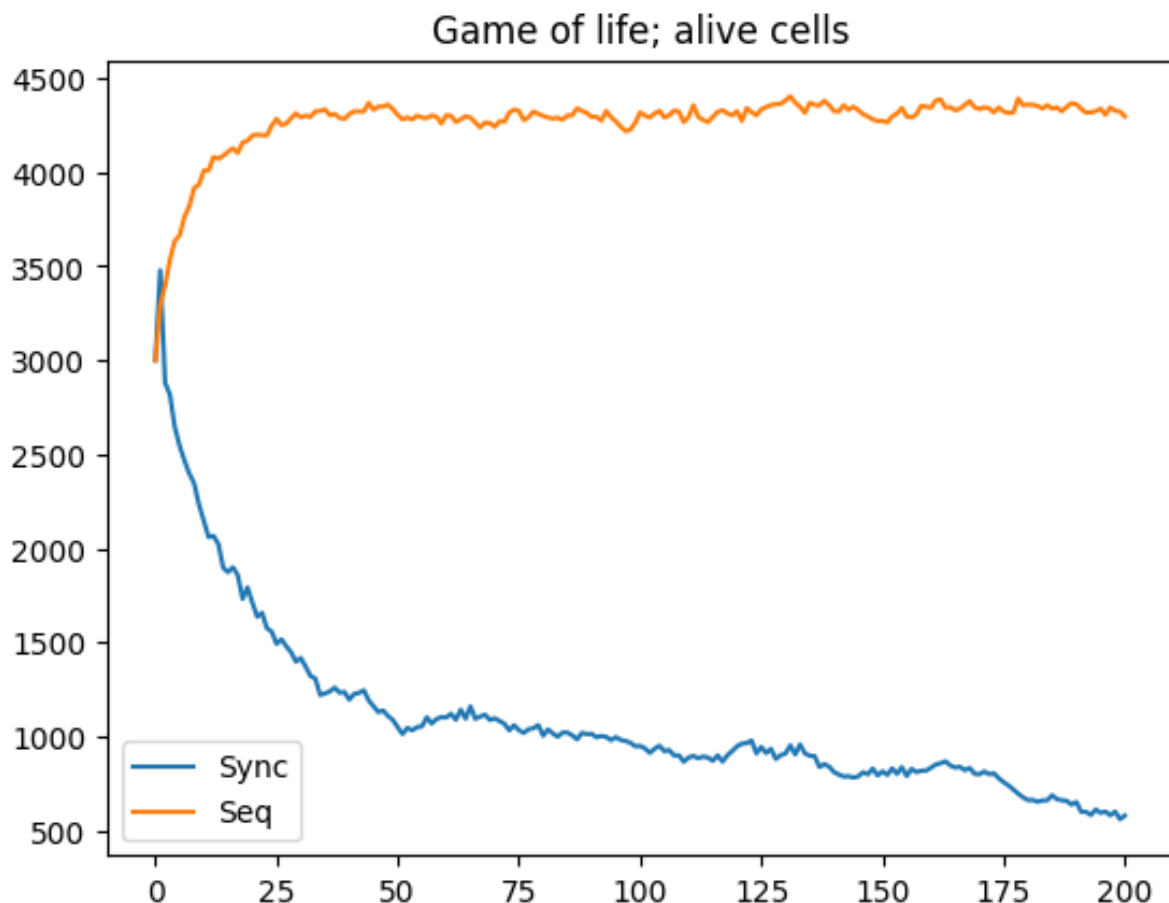


Figure 3: Alive cells - sequential vs synchronous

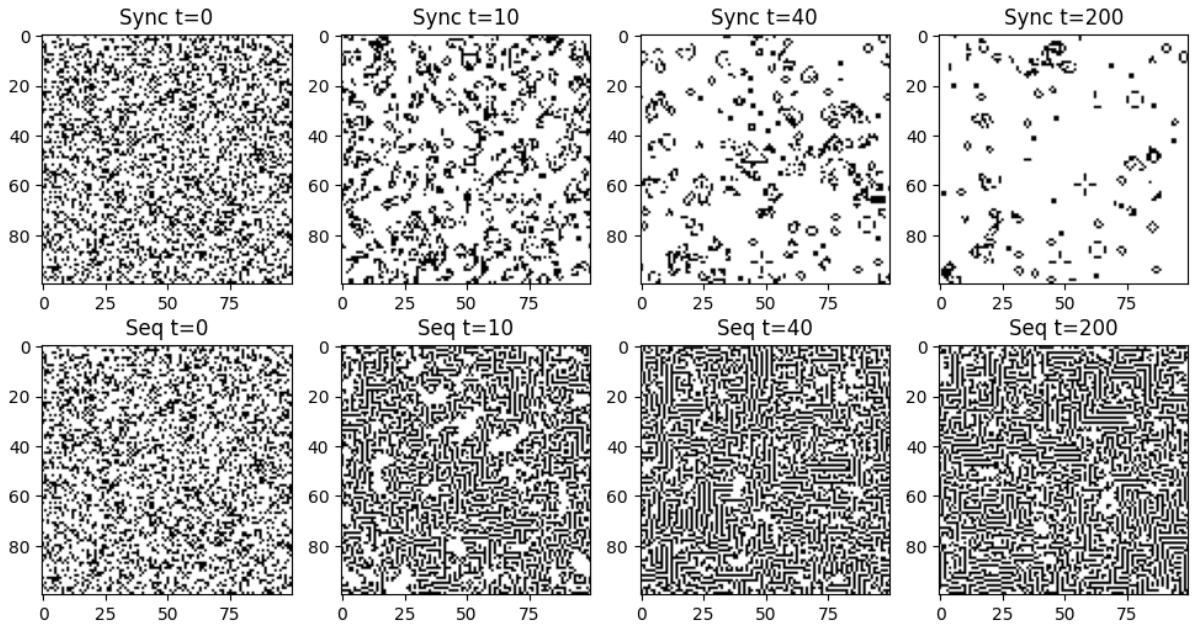


Figure 4: Snapshot plot

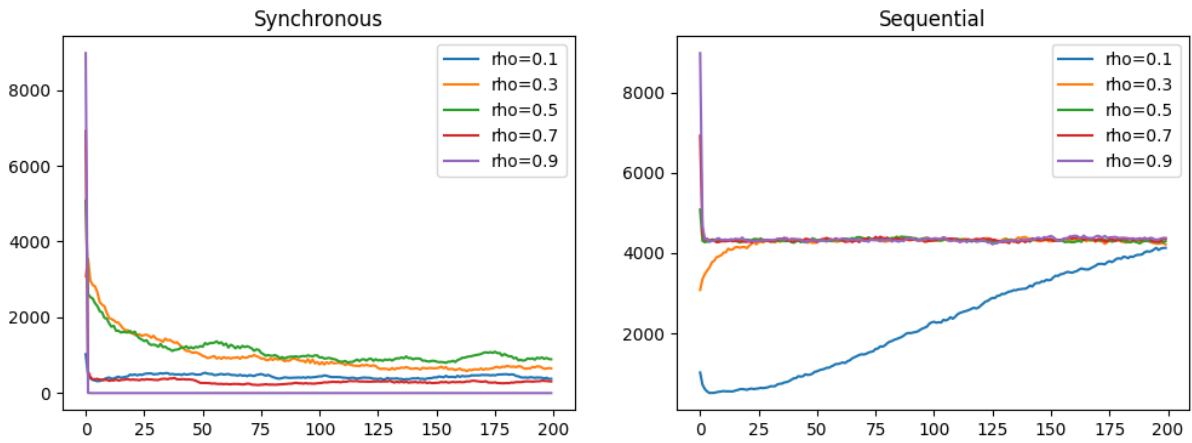


Figure 5: Plots visualizing different densities

2.3 Discussion

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- Why does the population stabilise at a higher level under sequential update?
- Compare the spatial texture of the final states.

”

In sequential updating, the population of Conway’s Game of Life tends to stabilize at a higher level because cells are updated one at a time rather than simultaneously, allowing each update to immediately influence the local neighborhood conditions of subsequent cells. As a result, situations in which large groups of cells would normally die simultaneously under synchronous updating are partially avoided, since the death or survival of one cell can alter the environment in ways that preserve neighboring cells. The spatial texture of the final state also differs significantly between the two variants:

- synchronous updating commonly produces recognizable dynamic structures such as oscillators and gliders, whose behavior depends on precise timing across the entire grid
- sequential updating disrupts this coordinated evolution and instead generates clumped, frozen patterns of inactive cells that lack the organized motion and recurring structures characteristic of synchronous dynamics

3 Day and Night Automation

3.1 Definition

Day & Night cellular automaton is a two-dimensional, two-state cellular automaton in the same family as Conway’s Game of Life, defined by a local, outer-totalistic rule in which each cell updates based on the number of live neighbors it has, but with a distinctive symmetry: patterns evolve in a way that is invariant under swapping live and dead states, meaning structures that form in one “polarity” have corresponding behavior when inverted. Its rule is typically expressed as B3678/S34678, indicating the neighbor counts that cause a dead cell to become alive and a live cell to survive, and this structure produces a rich mix of stable formations, oscillators, and mobile patterns despite its simple definition. The automaton is notable because this duality between states leads to visually striking dynamics in which dense regions and sparse regions can play interchangeable roles, often generating intricate, self-organizing behavior that sits between ordered stability and chaotic growth.

3.2 Discussion

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- Do both schemes respect the dead/alive symmetry of the rule (population near 50%)?
- Which scheme shows oscillations around the equilibrium? Why?
- Which converges faster to a stationary state?
- Compare the texture
- For whom does this update change affect, and why?

”

In the Day & Night cellular automaton, the preservation of the dead–alive symmetry depends strongly on the update scheme:

- synchronous updating respects this symmetry well, typically maintaining an approximate balance between states and keeping the system near the 50% occupancy level
- sequential updating breaks the symmetry by allowing intermediate state changes to bias subsequent updates, causing the system to drift away from this balance

In terms of temporal behavior, synchronous evolution produces strong oscillations, as large coherent regions can flip between high and low density states in coordinated cycles (very similar to natural day and night cycle), while sequential updating suppresses these global oscillations because updates promote locally and irregularly rather than across the entire grid at once. This difference also affects convergence, as sequential updating tends to converge more quickly to a stable configuration, since the staggered updates rapidly lock the system into a disordered, locally consistent state, whereas synchronous updating can sustain long dynamical cycles without settling. Visually, these dynamics produce distinct textures, with synchronous evolution appearing as structured blobs of light and dark that continuously trade places, while sequential evolution results in a more uniformly frozen, noise-like pattern lacking big motion.

3.3 Results

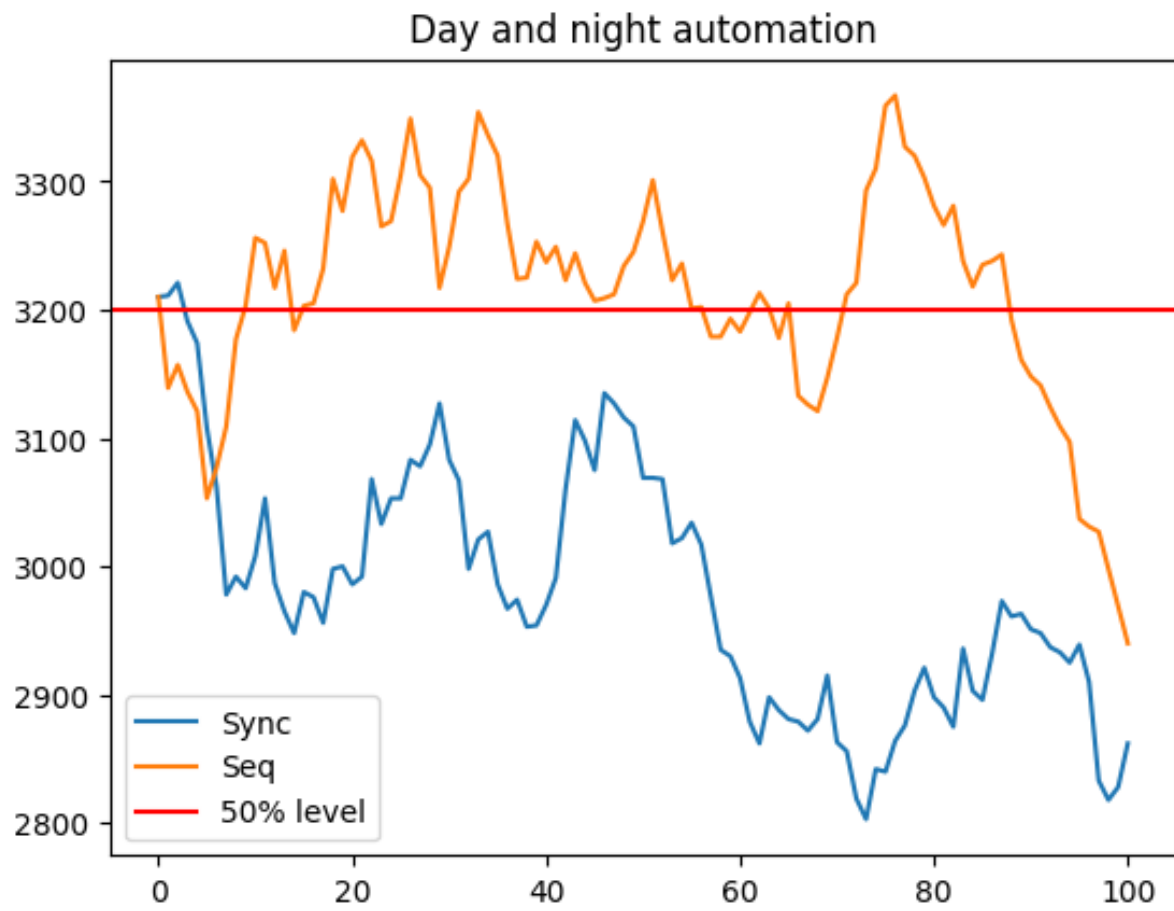


Figure 6: Time series plot

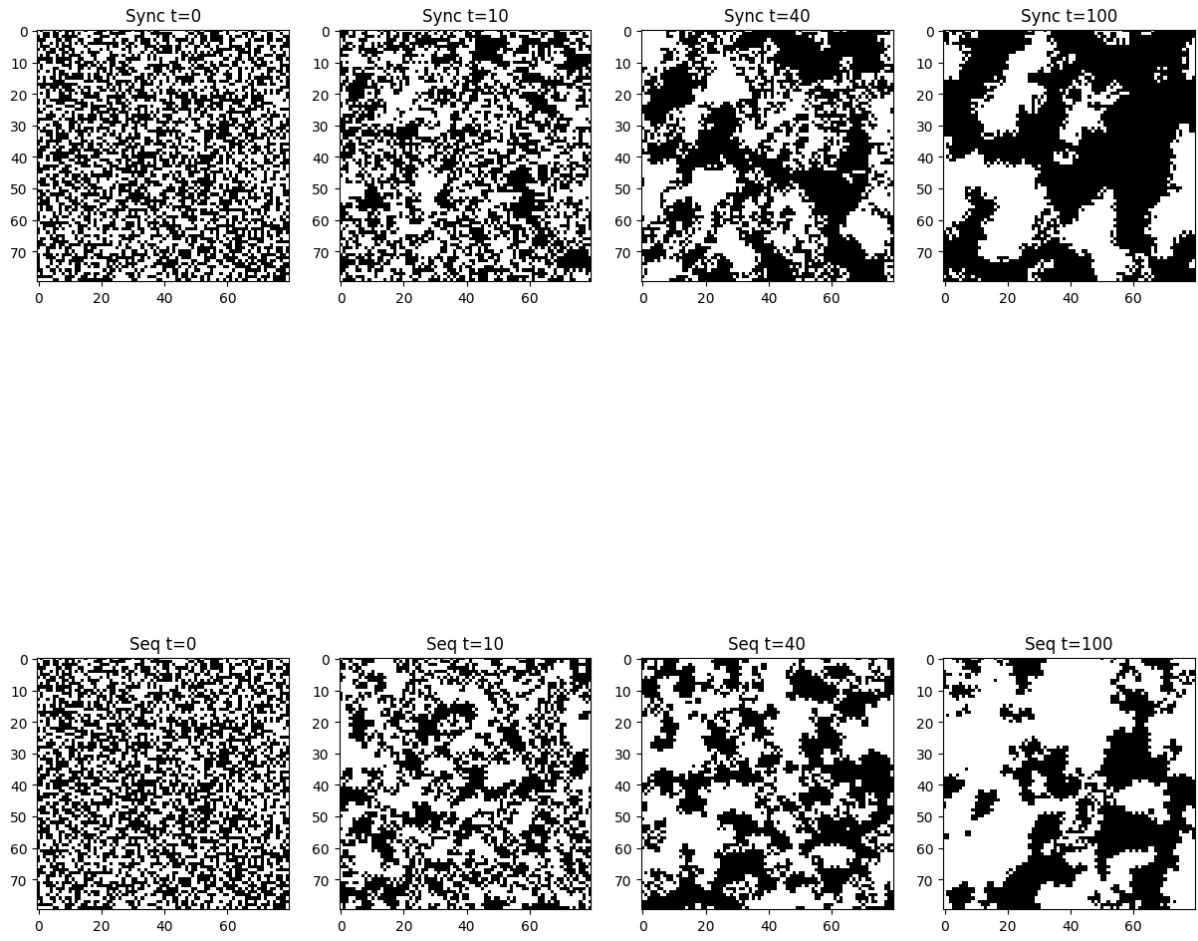


Figure 7: Snapshot plot